

HAMDI CAN ERNALBANTOĞULLARI

SOFTWARE TEST ENGINEER

QA & TEST AUTOMATION | AI-ASSISTED TESTING | TECHNICAL PROJECT COORDINATION

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PROFESSIONAL SUMMARY

Software Test Engineer with 4+ years of engineering experience in software testing and electronic system design. Spent 3 years in testing and quality assurance at Vestel R&D. Worked on 9 major Smart TV software projects and 20+ sub-projects, handled 4,000+ defect records, and investigated 150+ production and field issues. Work also covered release testing, system and requirements analysis, and coordination across teams. Developed test automation skills through independent projects and used AI tools for code analysis, debugging, test scenario design, and documentation. Career interests include test automation, AI-assisted testing, system analysis, and technical project management.

CORE COMPETENCIES

Software Testing & QA | Test Automation | Python | Pytest | Playwright | Selenium | API Testing | AI-Assisted Testing | System Analysis | Requirements Analysis | Technical Project Coordination | Release Validation | Defect Management | Jira | CI/CD | GitHub Actions

PROFESSIONAL EXPERIENCE

Software Test Engineer - Vestel Electronics Inc.

November 2022 – November 2025
Manisa, Türkiye | Hybrid

- Conducted software testing and quality assurance activities across 9 major Smart TV projects and more than 20 panel/RAM configuration sub-projects.
- Contributed to 3 new product development projects, 5 maintenance projects, and 1 finalized project.
- Designed and executed functional, regression, smoke, sanity, exploratory, and release validation tests.
- Planned and performed an average of 2 software release testing cycles per week.
- Developed test scenarios and test cases based on functional requirements, technical specifications, edge cases, and negative testing conditions.
- Worked with over 4,000 defect records using Jira, including defect reproduction, investigation, verification, regression testing, and closure.
- Investigated more than 150 production and field issues, including manufacturing interruptions and customer-reported software defects.
- Analyzed software logs, system behavior, build versions, and technical evidence to support root cause investigations.
- Prepared comprehensive defect reports detailing reproduction steps, expected and actual results, severity, priority, and release impact.
- Collaborated with software development, hardware engineering, production, quality control, and customer support teams.
- Supported release readiness evaluations, technical problem-solving, requirements analysis, and quality risk assessments.
- Contributed to cross-functional coordination and communication throughout testing and release processes.

Key Technologies & Skills: Jira, Software Testing, QA, Functional Testing, Regression Testing, Release Validation, Defect Management, System Analysis, Requirements Analysis, Technical Coordination

System Design Engineer - Vestel Electronics Inc.

Manisa, Türkiye | Hybrid
July 2021 – November 2022

- Designed electronic systems for LED display modules and cabinet structures.
- Analyzed technical specifications, system requirements, and hardware design constraints.
- Supported system-level design verification, technical validation, and product development activities.
- Contributed to requirements analysis, technical documentation, and engineering problem-solving.
- Participated in hardware-software interaction analysis and system design activities.
- Collaborated with multidisciplinary engineering teams to support product development and technical coordination.
- Supported technical project coordination and stakeholder communication.

Key Technologies & Skills: System Design, Electrical Engineering, Electronics Engineering, Requirements Analysis, Technical Project Coordination, System Verification, MATLAB, Simulink

PROFESSIONAL INTERESTS

Software Quality Engineering | Test Automation | AI-Assisted Testing | System Analysis | Technical Project Management | Continuous Improvement

TECHNICAL PROJECTS

Python Test Automation Portfolio

Python | Pytest | Playwright | Selenium WebDriver | REST APIs | Postman | GitHub Actions

- Developed a Python-based UI and API test automation portfolio.
- Automated 15 documented UI test scenarios across TodoMVC and SauceDemo using Playwright and Pytest.
- Implemented the Page Object Model (POM), reusable fixtures, and smoke/regression test markers.
- Developed Selenium WebDriver automation for SauceDemo workflows.
- Created API automation using Python Requests, Pytest, and REST APIs.
- Implemented reusable API client layers, JSON handling, and request/response validation.
- Configured 3 GitHub Actions workflows for automated testing, Newman execution, and CodeQL analysis.
- Integrated JUnit XML reports, screenshots, logs, and Playwright traces for failure investigation.

WasteWisely – Digital Waste Auditor

Python | FastAPI | Pytest | Playwright | pywebview | Generative AI

- Developed a Windows desktop application for directory scanning, digital waste classification, sensitive-file detection, cleanup, and archiving.
- Built application components using Python and FastAPI.
- Implemented automated tests using Pytest and Playwright.
- Performed manual testing, workflow validation, and application packaging verification.
- Used Codex-assisted repository analysis and debugging with human verification.
- Applied AI-assisted engineering techniques to support technical investigation and development.

OKX Trading Bot

Python | Pytest | API Integration

- Developed a modular Python-based spot-trading application.
- Implemented technical scoring, market-regime detection, and risk management components.
- Developed trade sizing, reconciliation, Take Profit, and Stop Loss functionality.
- Integrated Telegram-based monitoring.
- Created Pytest-based automated tests covering trading execution, reconciliation, risk controls, market-regime behavior, and trade sizing.

TECHNICAL SKILLS

- **Software Testing & Quality Assurance:** Functional Testing, Regression Testing, Smoke Testing, Sanity Testing, Exploratory Testing, Test Case Design, Test Scenario Design, System Testing, Real-Device Testing, Release Validation, Defect Lifecycle Management, STLC
- **Test Automation:** Python, Pytest, Playwright, Selenium WebDriver, Page Object Model (POM), Pytest Fixtures, UI Automation, Automated Regression Testing, JUnit XML Reporting, Playwright Trace Viewer
- **API Testing:** REST APIs, Postman, Newman, Python Requests, JSON, API Client Layers, Request/Response Validation
- **Artificial Intelligence & LLMs:** Generative AI, AI-Assisted Testing, Large Language Models, GPT, Claude Code, Gemini, Codex, AI-Assisted Debugging, Codebase Analysis, Test Scenario Generation, Test Coverage Exploration
- **System Analysis & Project Coordination:** Requirements Analysis, System Analysis, Test Analysis, Technical Project Coordination, Release Coordination, Stakeholder Communication, Cross-Functional Collaboration, Quality Risk Assessment, Technical Documentation, Root Cause Investigation
- **Development & CI/CD:** Git, GitHub, GitHub Actions, CI/CD, Chrome DevTools, FastAPI, SDLC
- **Engineering Tools:** MATLAB, Simulink

CERTIFICATIONS & PROFESSIONAL TRAINING

- **Artificial Intelligence Specialization Training - Opsiyon Akademi / Riscon Academy**

Artificial Intelligence | Python

- **ISTQB 2018 Foundation Level (CTFL) Exam Preparation - Udemy**

Software Testing | Manual Testing | Test Management

- **Generative AI – GPT, Gemini, Claude & MCP - Udemy**

Generative AI | LLMs | GPT | Gemini | Claude | MCP | AI-Assisted Engineering

EDUCATION

Izmir Institute of Technology
Bachelor of Science (BSc) – Electrical and Electronics Engineering
Electronics and Communication Program
September 2015 – March 2021

LANGUAGES

Turkish – Native
English – Professional Working Proficiency (B2)
Italian – Elementary (CILS A1)

REFERENCES - Available upon request.